

### Question 1:

Sector: categorical & nominal

Market Cap: scalar & continuous

Daily Returns: scalar & continuous

### Question 2:

|             | Sector  | MC         | Ret. |
|-------------|---------|------------|------|
| Mode        | Finance | x          | x    |
| Median      | x       | 495.3      | 0.32 |
| Mean        | x       | 930.8      | 0.53 |
| #categories | 5       | x          | x    |
| Minimum     | x       | 125.7      | -1.1 |
| Maximum     | x       | 2,850.5    | 3.45 |
| Range       | x       | 2,724.8    | 4.55 |
| Variance    | x       | 751,409.98 | 1.48 |
| Std dev     | x       | 866.84     | 1.22 |
| 1st Q       | x       | 350.35     | -0.4 |
| 3rd Q       | x       | 1,300.6    | 1.2  |
| IQR         | x       | 950.25     | 1.6  |

Q2 cont.

## Market Capitalization

◦ Median 13 obs  $(13+1)/2 = 7$

|       |       |     |       |                                                                   |       |     |        |
|-------|-------|-----|-------|-------------------------------------------------------------------|-------|-----|--------|
| 125.7 | 155.2 | --- | 485.8 | <span style="border: 1px solid black; padding: 2px;">495.3</span> | 720.4 | --- | 2850.5 |
| obs 1 | 2     |     | 6     | 7                                                                 | 8     |     | 13     |

median

◦ Mean,  $\mu$

$$\mu = \frac{125.7 + 155.2 + \dots + 2850.5}{13} = 930.08$$

◦ Range = Maximum - minimum =

$$= 2850.5 - 125.7$$
$$= 2724.8$$

◦ Variance,  $\sigma^2$

$$\sigma^2 = \frac{(125.7 - \mu)^2 + (155.2 - \mu)^2 + \dots + (2850.5 - \mu)^2}{13}$$

$\mu = 930.08$

$$= 751,409.98$$

◦ Standard deviation

$$= \sqrt{\text{variance}} = \sigma = \sqrt{751409.98} = 866.84$$

Q2 cont.

• IQR

1<sup>st</sup> 2<sup>nd</sup> 3<sup>rd</sup> 4<sup>th</sup> 5<sup>th</sup> 6<sup>th</sup>  
 125.7 152.2 280.6 420.1 475.1 485.8

7<sup>th</sup>  
 495.3  
 Median

8<sup>th</sup> 9<sup>th</sup>  
 720.4 770.9

10<sup>th</sup> 11<sup>th</sup> 12<sup>th</sup> 13<sup>th</sup>  
 1150.9 1450.3 2710.2 2850.1

$$1^{st} Q = \frac{280.6 + 420.1}{2} = 350.35$$

$$3^{rd} Q = \frac{1150.9 + 1450.3}{2} = 1300.6$$

$$\Rightarrow IQR = 3^{rd} Q - 1^{st} Q =$$

$$= 1300.6 - 350.35 =$$

$$= 950.25$$